

Xinwei Zhuang

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RESEARCH EMPHASIS

Energy and Environment: AI for sustainable energy transition; Building stock analysis; Urban building energy modeling; Energy conservation and projection; Ventilation and daylighting optimization

Computation: Machine learning; Data science; Network science; Computer vision; Computer graphics

EDUCATION

University of California, Berkeley Ph.D. in Architecture	Berkeley, United States 2020-2025
The Bartlett School of Architecture, University College London M.S. in Architectural Computation	London, United Kingdom 2016-2017
Edinburgh Napier University B.S. in Civil Engineering	Edinburgh, United Kingdom 2012-2016

EXPERIENCE

Texas A&M University <i>Assistant Professor</i>	College Station, United States 2025 – Present
Massachusetts Institute of Technology <i>IvyPlus Exchange Scholar, Department of Urban Studies and Planning</i>	Cambridge, United States 2024 – 2025
Lawrence Berkeley National Lab <i>Research assistant</i>	Berkeley, United States 2022 – 2025
- Decarbonizing Energy through Collaborative Analysis Routes and Benefits (DECARB) - Multi-scale Energy Conservation Measurement (ECM) evaluation and projection for U.S. residential and commercial building sectors, provide policy insight for ECMs and regulations	
Architecture Studio <i>Architectural designer</i>	Shanghai, China 2018 – 2020
China Shanghai Architectural Design & Research Institute Co. Ltd <i>Research assistant</i>	Shanghai, China 2017 – 2018

INVITED TALK

- 2026** Block-Scale Solar PV and Battery Storage Co-Optimization for Energy Resilience
ASHREA summer conference, Austin, TX, USA
- 2026** Community-driven energy resilience with decentralized networks
COA Showcase, Texas A&M University, TX, USA
- 2025** When the city goes dark: Measuring building vacancy based on energy consumption in Rio de Janeiro
Presentation at MIT Energy Conference 2025, Boston, MA, USA
- 2024** The role of architecture in the urban energy landscape
Guest lecture at the Manchester Urban Institute, University of Manchester, UK
- 2024** Decomposing and recomposing of the urban energy environment with building archetype and urban building energy modeling
Guest lecture at Foster + Partner, UK
- 2023** Performance-driven architectural design with deep learning
Invited talk at 33rd Young Scholar Forum, Nanjing University, China

- 2026** PI COA Small Grant (**\$20k**)
Community-Centered Energy-Sharing Neighborhood for Texas Triangle
- 2025** Co-PI Targeted Proposal Teams (**\$50k**)
Multi-Source Integration of New Satellites, Low-Cost Sensors, and Human Mobility for Precision Air Pollution Exposure Assessment
- 2024** Co-PI Lau Grants for Just Climate Futures (**\$35k**)
Energy-Sharing Neighborhood: Community-Driven Energy Resilience with Decentralized Networks
- 2024** PI Hearts to Humanity Eternal (**\$10k**)
Urban Synergy: Performance-Driven Urban Neighborhood Modeling with Hierarchical Networks for Energy Resilience

SELECTED PUBLICATION

* equal contribution

1. **Zhuang X.** and Tan H. (2026) Automated Neighborhood Archetype: Integrating Morphology and Topology for Urban Building Energy Modeling, *Energy and Buildings*. Vol. 362, Article 117504. DOI: [10.1016/j.enbuild.2026.117504](https://doi.org/10.1016/j.enbuild.2026.117504)
2. **Zhuang X.**, Lyv G., Zhao Z. and Caldas. L (2025) Rapid Assessment of Solar Potential for Building Surfaces in Complex Urban Morphologies Based on Vector Processing, *Solar Energy*, Volume 294, 2025, 113482, ISSN 0038-092X, DOI: [10.1016/j.solener.2025.113482](https://doi.org/10.1016/j.solener.2025.113482)
3. **Zhuang X.**, Zhu P., Yang A. and Caldas L. (2025) Machine Learning for Generative Architectural Design: Advancements, Opportunities, and Challenges, *Automation in Construction*, Volume 174, 2025, 106129, ISSN 0926-5805, DOI: [10.1016/j.autcon.2025.106129](https://doi.org/10.1016/j.autcon.2025.106129).
4. Chen X., Lyv G., **Zhuang X.**, Duarte C. and Schiavon S. (2025) Integrating Symbolic Neural Networks for Advanced Modeling in Building Physics: A Study and Proposal. *Journal of Building Engineering*, Vol. 111, 113033, <https://doi.org/10.1016/j.jobbe.2025.113033>
5. **Zhuang X.**, Chu X., Liang, J., Zhu P., Gonzalez, M. and Caldas, L. (2024) Across scales: Hierarchical Urban Graph for Neighborhoods Partition and Decentralized Energy Autonomy, in *2024 Conference on Association for Computer Aided Design in Architecture (ACADIA)*, Calgary, Canada. [paper]
6. **Zhuang X.***, Huang Z.*, Zeng W., and Caldas L. (2023) MARL: Multi-scale Archetype Representation Learning for Urban Building Energy Modeling, workshop on Computer Vision Aided Architectural Design (CVAAD), at *International Conference on Computer Vision (ICCV)*, Paris, France. DOI: [10.1109/ICCVW60793.2023.00171](https://doi.org/10.1109/ICCVW60793.2023.00171)
7. **Zhuang X.***, Huang Z.*, Zeng W., and Caldas L. (2023) Encoding Urban Ecologies: Automated Building Archetype Generation through Self-Supervised Learning for Energy Modeling, in *2023 Conference on Association for Computer Aided Design in Architecture (ACADIA)*, Denver, United States of America. papers.cumincad.org/cgi-bin/works/paper/acadia23_v2_532
8. **Zhuang X.**, Luo, N., Hong T. and Koenig, M. (2023) What can we learn from Honda Smart Home with high-resolution monitored performance data: A zero-net energy home in California, in *18th International IBPSA Conference and Exhibition* In Shanghai, China. DOI: [10.26868/25222708.2023.1315](https://doi.org/10.26868/25222708.2023.1315)
9. **Zhuang X.**, Ju Y., Yang A. and Caldas L. (2023) Synthesis and Generation for 3D Architecture Volume with Generative Modeling, in *International Journal of Architectural Computing*, Vol. 21, Issue 2: AI, Architecture, Accessibility, & Data Justice. DOI: [10.1177/14780771231168233](https://doi.org/10.1177/14780771231168233)